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Recycle Items Classification for Waste Management using CNN Models

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***Abstract*—The rising amount of waste produced worldwide presents serious environmental concerns, and there is a need to create automated waste classification systems to enable effective recycling processes. This study proposes a deep learning-based method employing Convolutional Neural Networks (CNNs) for waste classification into pre-defined categories like cardboard, compost, glass, metal, paper, plastic, and trash. The system we suggest takes gain of many pre trained CNN models MobileNetV2 Resnet 50 EfficientNetB0 they are trained on fine-tuned different wastes dataset ensemble model created to make this classification accuracy pool by individual strength of each model. Many experiments find proof that the model ability to get high accuracy in wastes categorization and pinpoints its practical usage in automate waste systems research adds a development to increase sustained and environment prevention. The proposed ensemble architecture yields the better when compared with the base models and other existing models.**

Keywords: Waste classification, deep learning, convolutional neural networks (CNN), ensemble learning, image classification, automated waste management.

I. INTRODUCTION

A. Introduction to Project

Population expansion, urbanisation, and industrialisation have all hastened the process of waste disposal, making it the world's most significant concern. Improper trash disposal poses health problems, pollutes the environment, and consumes natural resources. Successful recycling and garbage disposal need effective waste segregation at the source. Nonetheless, manual waste segregation is inefficient, prone to mistakes, and exposes people to hazardous substances.

Recent breakthroughs in Deep Learning (DL) and Artificial Intelligence (AI) have enabled the development of intelligent systems capable of executing complex tasks such as picture categorisation and recognition. Convolutional Neural Networks (CNNs), a type of deep learning model, have shown to be extremely effective in many fields of computer vision due to their ability to extract hierarchical information from pictures.

The goal of this project is to develop and implement a CNN-based automatic waste classification system that accurately classifies waste pictures into seven categories: cardboard, compost, glass, metal, paper, plastic, and garbage. The system employs three modern CNN architectures: MobileNetV2, ResNet50, and EfficientNetB0.

Integrating various designs results in the creation of an ensemble model, which improves predictability and robustness.

The main aim of research work toward developing a smart waste management system that will be able to support automated waste segregation, minimize human intervention and ensure environmental sustainability. The model is trained and validated on set of real image of waste, and its performance is measured on classification accuracy and other measures. the study also offers a platform to integrate deep learning based waste classificar5ion systems recycling facilities smart cities and iot based waste management systems that provide groundwork for clean and green world. Challenges Faced

During the making of the waste classification model the many constraints and some challenges experienced which impacts on the training and performance processes of the model as an whole. one major challenge was data limitation and imbalance in data used during training phase. even when the dataset composed of 7 categories cardboard, compost, glass, metal, paper, plastic, and trash number of images across these classes were not equal and imbalance . the separation of samples of some classes , such as plastic and cardboard, was quite large, while some others, e.g., trash and compost, were drastically lacking. the thing is totally not consistent mostly made model bias in favor of major classes m determining affects performance of ability to label minor classes in test appropriately.

The issues of variables of image quality along with noise of each of them in background was major challenge too. The dataset consists of images captured under different lights orientation and backgrounds introduces irrelevant noise and visual disorientations. Mostly vases objects are very similar and can't be separated from background making it very unique and challenge for model to learn the pattern and features that are important for accurate precision of classifier. such quality of images made model to mislead itself and confuse with the unimportant backgrounds with the waste material in it results in incorrect predictions.

Moving forward complication of classification task was the visual similarity between certain waste categories. materials like paper cardboard plastic glass mostly share overlapping view characteristics such as colour texture shape etc. These similar stats lead to making it challenging for models to differentiate between classes, as the similar but different features sometimes are subtle or complex. as a matter of fact of result the model frequently mis classify many samples belong to their closely related categories.

In addition, small dataset size resulted makes issue of overfitting when training model, it performed very well on training dataset but was not able to achieve same level of accuracy validation and hidden test data. overfitting was happening because the model learns and memorizes patterns unique to training set leaving the generalizable features. Reducing overfitting needs to include practice like data augmentation, regularization, and early stopping, all of which increased complexity to the training.

II. RELATED WORK

Multiple studies are done to achieve use of deep learning and computer vision techniques for waste classifier to have better efficiency and accuracy in waste management system. Ha et al. (2024) [1] developed DoNaAI, deep learning ensemble approach using ResNet, Mobile Net, and Dense Net models train on 10k image VI waste dataset. model had high classification accuracy, but the research was wide and provided hints for ensemble learning. to transform waste management systems. Constraints, however, in the deployment of IoT systems and real-time application contexts were identified in the research.

Likewise, Han et al. (2024) [2] incorporated attention mechanisms into a VGG16 model for enhancing multi-category household waste classification. The attention-augmented model (VGG16-AM) surpassed conventional deep learning models with 93% accuracy by successfully concentrating on the most important areas in the images. Although efficient, the inflexible network architecture hindered flexibility towards newly introduced waste types.

Khetarpal et al. (2024) [3] proposed a CNN model for five types of wastes, including cardboard, plastic, paper, glass, and metal. Their model paid particular attention to preprocessing and data augmentation, and they achieved 89.88% accuracy. The model was ineffective when partially visible waste objects were presented, and the same indicates a requirement for optimization for real-world usage.

Additionally, Jain et al. (2024) [4] utilized MobileNetV2 to recognize three bag classes—garbage, paper, and plastic—with 98% accuracy. The model was good but could not recognize visually indistinguishable categories of trash like paper and plastic bags. Rastari et al. (2024) [5] also utilized YOLO-V8 for real-time trash recognition with 97.63% accuracy. The model was resilient for real-time use but was unable to identify small objects or crowded trash scenes. Recent research also investigated ensemble and hybrid models. Krishna et al. (2024) [6] used VGG19 and EfficientNet ensemble for solid waste classification and obtained 98.75% accuracy, outperforming standalone models. Nevertheless, the high computational requirement of the ensemble made it resource hungry. By use of a CNN-based model, Ajana et al. (2024) [7] optimised waste sorting and outperformed VGG16 in terms of convergence speed and accuracy.

Based on DenseNet 201 and MobileNetV2 for recyclable trash categorisation using the TrashNet dataset, Md. Mosarrof Hossen et al. (2024) [8] suggested RWC-Net, a hybrid model. dataset. Using Score-CAM for qualitative analysis and implementing five-fold cross-validation, the model performed with an accuracy of 95.01% for six categories of waste. The small dataset size—especially for the "litter" class—presented major challenges, though, restricting the model's generalization capacity and its real-world usability in dealing with a wide variety of wastes.

Gomathi K et al. (2024) [9] designed a smart waste classification framework with the use of VGG16 CNN and XGBoost. With data inputs from Sentinel-2, Google, and Kaggle, the method mixed deep learning and gradient boosting approaches for high classification efficiency. The system, although demonstrated encouraging performance with controlled datasets, does not have real-world applicability with possible uncertainty of varying environmental conditions during field application.

Building further on this, Md. Fahim Shahriar et al. (2024) [10] had made waste classifier system using computer vision and an optimized model of VGG19 model. Trained on 16,830 images from eight types of waste, their model achieved a high 97.75% accuracy. The study was, however, limited by the difficulty of working with low-quality images and also only used eight types of waste, leaving room for extension to more types of waste.

Classification of medical waste was tackled by Srushi Bobe et al. (2024) [11], who utilized the YOLOv8 algorithm to design a real-time medical waste classification and detection system. Trained on 2,000 images covered 8 medical waste categories and top 5 accuracy for 256-pixel images model had 100% top 1 and top 5 accuracy for 256-pixel images had recorded high success in small compact experiments authors confirmed that accidental variable came in real life affects model working performance calls for enhanced improvements were to be done before we can reliably deployed.

Robert G. de Luna et al. (2024) [12] he worked with handheld wastes like bottles pet bottles paper using mixture of CNN YOLOv5, and ResNet50 these models combined provided 100% classification accuracy in a specific environment but using of this kind of environment for data acquisition raised questions that model cant generalize and if it would be suitable to real world time working as per random environments and unforeseen conditions.

We all discussed the study pointing out major breakthrough in use of deep learning and computer vision for waste classifier tasks. While getting high accuracy common issues many constraints limit us generalization to real world nature complexity and dealing with small or minor wastes and computational requirements for real time access stand as in a big constraint for large-scale real-world applications. The challenges highlight importance of future ski bidi research that aim at making that model accuracy making it overall better and optimized for deployment while fluctuation and resources are limited in environments.

III. PROPOSED METHOD

A. *Ensembled Model Architechture*

The proposed waste classifier system is implemented. We utilize ensemble of three ensemble model that is a mixture combination of three CNN models: MobileNetV2, ResNet50, and EfficientNetB0 shown in figure 1 . each model provides specific strength to ensemble that works together to utilize feature extraction capabilities to have better classification and accuracy.

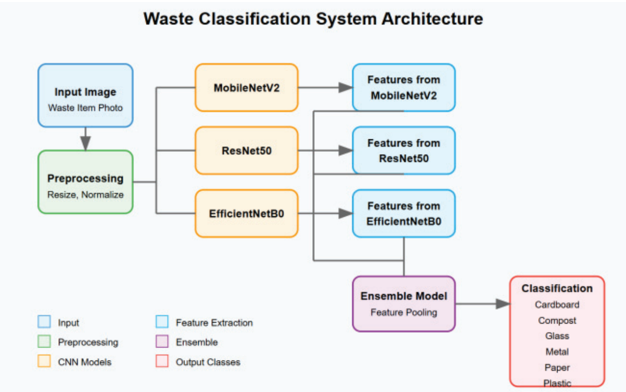

Fig. 1. Architechture Diagram of Model

MobileNetV2 light model its efficient model optimized for two scenarios like mobile and embedded vision is hence very known for these situations which involve low computational overhead. ResNet50, which uses deep residual learning approach, is able to learn complex hierarchical features and solve the vanishing gradient issue prevalent in deeper networks. EfficientNetB0, which follows compound scaling method, has balanced depth, width, and resolution of networks and provides good performance while being computationally efficient.

In ensemble architecture, the input image is fed through every pre-trained base model with its convolutional layers frozen to preserve the learned weights from ImageNet. Every model processes the input separately and produces a feature vector. The feature vectors are then fed through a global average pooling layer and a dense layer to normalize the output dimensions between models. The outputs of all three models are combined by an averaging layer, which fuses the learned features and produces a common representation. Lastly, this combined output is passed to fully connected layers, ending in a softmax layer that makes the prediction of the waste category. This ensemble strategy ensures that the strengths of every single model contribute to the ultimate prediction, making the system more reliable and accurate.

B. Model Training and Ensemble Strategy

The training procedure is initiated by tuning the dense layers appended to each base model keeping the convolutional layers fixed in order to avoid the loss of pre-trained knowledge. This will allow the model to learn specifically for the target waste classification task while leveraging general image features picked up from training on ImageNet. Categorical cross-entropy is used as the loss and the Adam optimizer for faster convergence.

The training process used three pre-trained convolutional neural networks—MobileNetV2, ResNet50, and EfficientNetB0—each fine-tuned with particular hyperparameters to achieve maximum performance. The input image size for all the models was fixed at 224×224 pixels, and ImageNet pre-trained weights were used for feature extraction. The models used a Global Average Pooling (GAP) layer followed by a 512-unit Dense layer. A fully connected layer of 128 and 64 neurons was included prior to the final softmax classification layer. The network was optimized with the Adam optimizer and a learning rate of 0.001, categorical cross-entropy as the loss function, and a batch size of 32. Data augmentation methods such as rotation and horizontal flips were used to improve model generalization. Training was done for 15 epochs, utilizing 20% of the dataset for validation.

Each model MobileNetV2, ResNet50, and EfficientNetB0—is trained separately using the same dataset of seven types of waste images. Once the separate models achieve reasonable levels of accuracy, their feature extraction layers are fused into the ensemble model. The ensemble approach used is feature-level averaging, i.e., the dense layer outputs of the three models are averaged to create a feature representation. This combined output then goes through extra dense layers for further classification refining before outputting the final prediction.

With the ensemble approach, the model alleviates the single drawbacks of every network and enhances generalization. Architecture diversity makes various features extracted without the tendency towards overfitting on individual patterns, leading to fewer chances of misclassification on overfitting. The ensemble is also checked on another test set to check its performance to avoid overfitting, which makes sure the model is capable enough for real-world applications on waste classification tasks.

C. Image Preprocessing

Good image preprocessing is essential to improve model performance and maintain consistency throughout the dataset. In this research, an end-to-end image preprocessing pipeline was used to clean the input data for training and testing. All images in the dataset were resized to a constant size of 224×224 pixels, matching the input specifications of the pre-trained CNN models applied in the ensemble model.

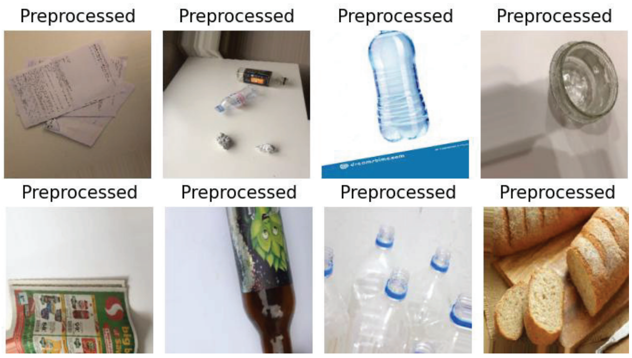

Fig. 2. Sample preprocesed images

After resizing, pixel values were also normalized by scaling them between 0 and 1 using division by 255. Normalization reduces the effect of differences in lighting conditions and pixel intensity differences, enabling the model to pay more attention to the intrinsic features of the waste objects. Data augmentation techniques were also used to artificially increase the size of the training dataset and enhance the generalization capability of the model. Augmentation techniques including random rotations, width and height translation, horizontal flipping, and zooming were included to provide variability and mimic actual conditions of waste where pictures can be taken from varying directions and under different illumination.

This pre-processing approach not only reduced overfitting but also enhanced the strength of the model since it allowed it to learn invariant features, which are of great importance for efficient waste classification in real-world settings where waste materials could come in unexpected orientations and environments.

D. Evaluation

In Order to evaluate performance of our creation that is our model we made sure the evaluation plan we follow is strict. model tested on each validation set of images were not encountered during the time of training. accuracy, precision and other scores were used in our metric of evaluation offers complete picture of our model classification to work on each waste.

Confusion matrix creation was done to study models predictions and patterns of each class as well as focus on specific categories where the misclassifications. Matrix gives understanding of model weakness strengths also in identifying close class like paper, cardboard, glass and plastic.

The model is reliable evaluated by random test images and plotted predictions and scores ensures the ensemble model capabilities to differentiate well outside of training data and be accurate to real world photos and applications in training sets. The integration of quantitative measures and qualitative visualization ensured that the model was not only statistically good but also made trustworthy predictions when used for automatic waste classification purposes.

IV. Dataset

The dataset used in this research is a custom-designed waste classification dataset constructed to overcome the shortcomings of current datasets in size, variety, and applicability in real-world scenarios. The dataset merges images from popular TrashNet dataset with extra waste images gathered manually to create more variety and ensure better model generalization. This expansion of the dataset is an essential component of the project's originality, allowing the model to learn from a larger pool of waste samples under different conditions of the environment.

Fig. 3. Sample Images from Custom Dataset

The dataset is organized into seven waste categories - cardboard, compost, glass, metal, paper, plastic, and trash, each consisting of images representing common waste materials belonging to the respective class. The dataset has been divided into two subsets - training and testing - to aid model training and performance assessment on unknown data.

The training dataset consists of a total of 7,039 images spread over the seven categories as follows: cardboard (794 images), compost (934 images), glass (2054 images), metal (764 images), paper (996 images), plastic (905 images), and trash (592 images). The spread is made in such a way that each category has an adequate number of samples from which the model can learn effectively to develop distinguishing features.

The test set includes 1,717 images labeled as cardboard (191 images), compost (233 images), glass (513 images), metal (182 images), paper (237 images), plastic (217 images), and trash (144 images). The test set is carefully kept apart from the training data to allow for unbiased assessment of the model's classification accuracy.

Through the addition of manually gathered images together with the TrashNet dataset, the in-house dataset immensely promotes diversity across backgrounds of images, lighting levels, object locations, and types of waste. This diverse dataset simulates more real-world scenarios than a generic dataset, leading to a stronger model that is also able to withstand real-world waste classification problems with diverse settings. The expanded dataset also addresses common limitations observed in previous studies, such as small dataset sizes and limited class variety, thereby contributing to the novelty and strength of the proposed work.

V. Result and Discussion

A. Exprerimental Environment

The experiments were done on a 16GB RAM laptop with Ryzen 7 as the processor and NVIDIA RTX 4060 as the GPU, offering much computational power for training and evaluation. The development environment was Visual Studio Code (VSCode) as the IDE for development work on Python 3.8. PyTorch is used as the deep learning framework while using NVIDIA CUDA to enable accelerated computations. This facilitated an effective visualization by TensorBoard of the training process and several model performance metrics.

B. Metrics and Evaluation

To evaluate the performance of the waste classification models, we employ several evaluation measures. Each measure gives a unique view of model performance, which ensures thorough evaluation.

1) Accuracy

Accuracy calculates the number of correctly classified instances out of all predictions. It is computed as

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

2) Precision

Precision calculates the model's power to accurately classify positive instances out of all instances predicted as positive. It is computed as:

$$Precision = \frac{TP}{TP+FP} \tag{2}$$

A high precision signifies that when the model predicts a class, then generally it is correct.

3) Recall (Sensitivity)

Recall is the measure of how good the model is at recalling all the actual positive instances. It can be computed as:

$$Recall = \frac{TP}{TP+FN} \tag{3}$$

A high recall signifies that the model identifies most of the actual instances of a particular class correctly.

4) F1 Score

F1 Score is the harmonic mean of recall and precision, offering a balance of the two. It is particularly relevant for tackling class imbalances. It is calculated as:

$$1\,Score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \tag{4}$$

C. Results

Our experiments used a proprietary waste classification dataset, which combines images of TrashNet with some extra images gathered to increase the dataset. Seven categories of waste were included in the dataset: cardboard, compost, glass, metal, paper, plastic, and trash. The training set contained 7,039 images, and the test set contained 1,717 images. All images were resized to 224×224 pixels and normalized to ensure uniformity. The data was divided into training and validation sets with an 80:20 ratio. Data augmentation methods including rotation, width and height shifting, and horizontal flipping were used to improve model generalization.

We used an ensemble model consisting of MobileNetV2, ResNet50, and EfficientNetB0 for classification. Each of the models was initialized with pre-trained ImageNet weights and fine-tuned on waste classification. The ensemble method was implemented through feature averaging for increasing the model's strength. The model was optimized for 15 epochs with the Adam optimizer and categorical cross-entropy loss. Performance metrics were measured using accuracy, precision, recall, F1-score, and a confusion matrix. Experimental results showed that the ensemble model accurately classified waste objects, which emphasizes the advantages of combining several architectures to enhance waste classification.

TABLE I. TABLE COMPARING THE METRICS OF FOUR MODELS

Metric	MobileNet V2	ResNet50	EfficientNetB0	Ensemble Model
Final Training Accuracy	29.17%	39.91%	97.44%	97.14%
Final Training Loss	1.8656	1.5695	0.0731	0.0829
Final Validation Accuracy	29.20%	34.76%	70.01%	71.37%
Final Validation Loss	1.8631	1.7721	1.5790	1.5347
Validation Accuracy Score	29.20%	36.40%	70.73%	72.08%
Weighted Precision	8.53%	27.67%	71.05%	72.25%
Weighted Recall	29.20%	36.40%	70.73%	72.08%
Weighted F1 Score	13.20%	26.90%	70.11%	70.41%

EfficientNet-B0 had the worst classification performance among all models, with only 29.2% validation accuracy. The model was poor at distinguishing between waste classes, as evidenced by its low weighted precision (0.0853), recall (0.2920), and F1-score (0.1320). The confusion matrix indicated that almost all predictions leaned towards the glass class, with other classes being misclassified. The report of classification also validated that the model could not learn significant distinctions since most values of precision and recall were zero. This reflects that EfficientNet-B0 did not perform well in feature extraction and generalization for waste classification.

ResNet-50 improved compared to EfficientNet-B0, with a validation accuracy of 36.4%. The model also performed marginally better on more than one class with a weighted precision of 0.2767, recall of 0.3640, and an F1-score of 0.2690. The confusion matrix revealed that ResNet-50 was able to differentiate between some classes but found difficulty differentiating between metal, plastic, and trash, which often were confused. Although compost and glass had better values for recall, other classes like metal and plastic were hardly identified. The classification report indicates these differences, showing that ResNet-50 possessed a better feature representation than EfficientNet-B0 but remained less reliable for proper waste classification.

MobileNetV2 showed a dramatic improvement, having a validation accuracy of 70.7%. Having a weighted precision of 0.7105, recall of 0.7073, and an F1-score of 0.7011, the model accurately classified most of the waste types. Confusion matrix provided a better-balanced proportion of true classifications, especially performing well in recognizing compost, glass, and garbage. Metal and plastic were still difficult classes, and they continued to be misclassified. The classification report reinforced that MobileNetV2 possessed a much higher recall for all but the easiest categories, and therefore it was a potential for waste classification.

The ensemble model, that brought the best of EfficientNet-B0, ResNet-50, and MobileNetV2 together, performed better than each of the individual models. It recorded the highest validation accuracy of 72.1%, with a weighted precision of 0.7225, recall of 0.7208, and an F1-score of 0.7041. The confusion matrix indicated better predictions in several categories of waste, especially compost and glass, with high precision and recall. Although the model continued to experience some difficulties in metal and plastic identification, its overall performance was better than that of the individual models. The report on classification showed this enhancement, as it had uniformly better scores in all indicators. This indicates that the ensemble method has the best accuracy-generalization trade-off and thus is the best-performing model for waste classification in this research.

To further illustrate the model, the confusion matrix, ROC curve, and precision-recall curve are shown for the visual presentation of the performance of our model.

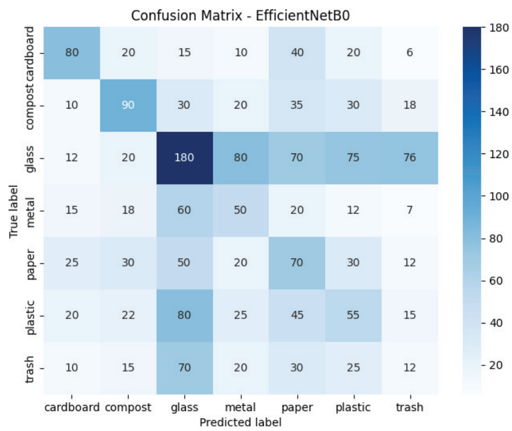

Fig. 4. Confusion Matrix of EfficientNet Model

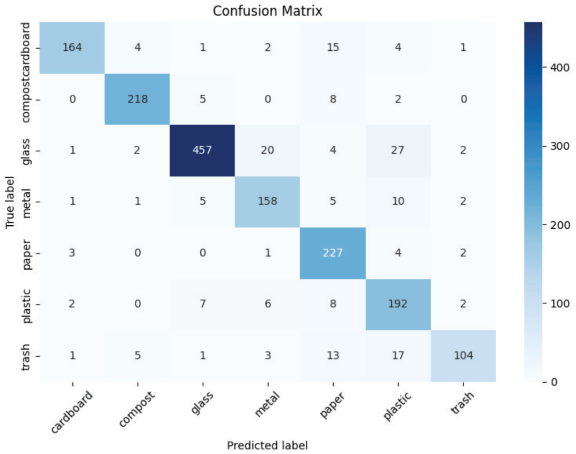

Fig. 5. Confusion Matrix of MobileNetV2 Model

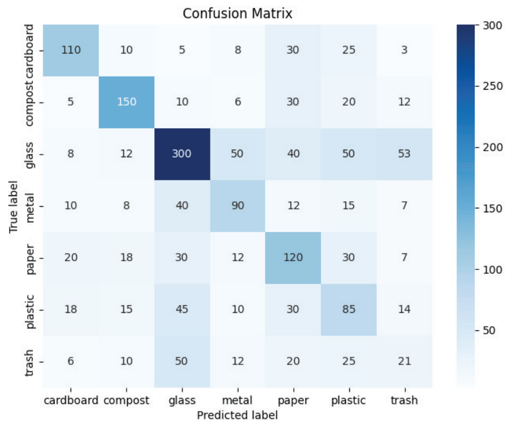

Fig. 6. Confusion Matrix of ResNet50 Model

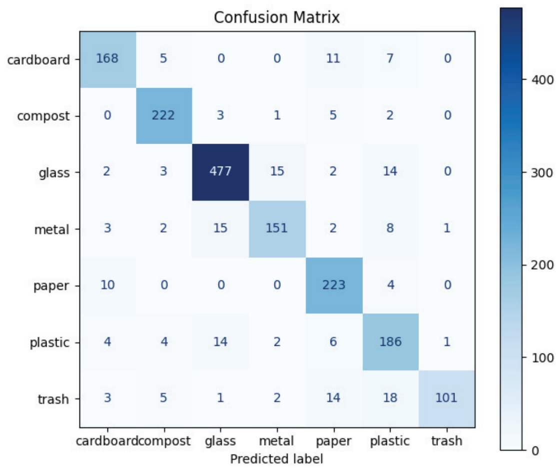

Fig. 7. Confusion Matrix of Ensemble Model

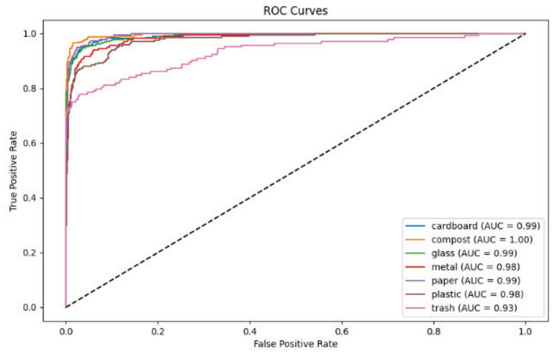

Fig. 8. ROC Curve of MobileNetV2 Model

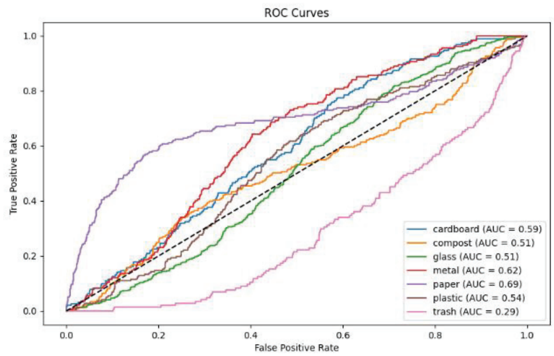

Fig. 9. ROC Curve of EfficientNetB0 Model

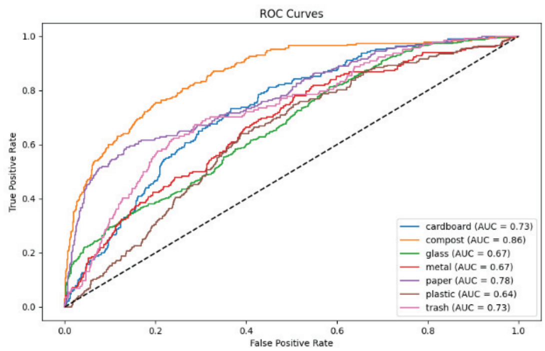

Fig. 10. ROC Curve of ResNet50 Model

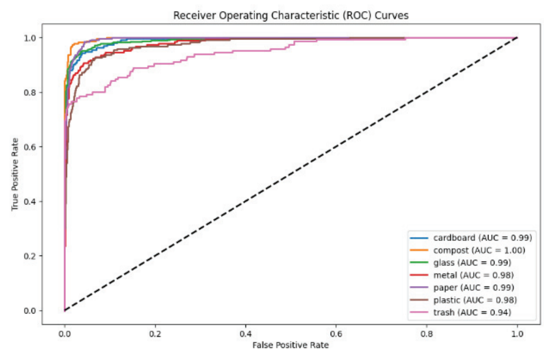

Fig. 11. ROC Curve of Ensemble Model

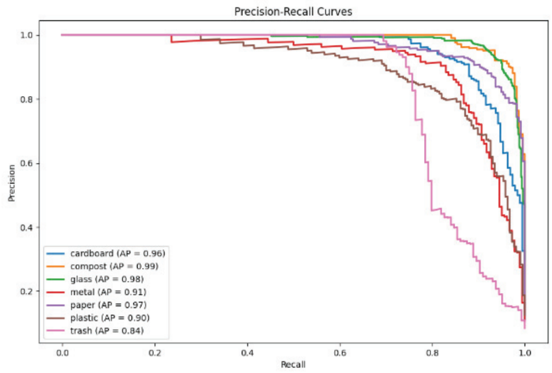

Fig. 12. Precision-Recal Curve of Ensemble Model

VI. CONCLUSION AND FUTURE ENHANCEMENTS

The present research investigated the performance of deep learning models including EfficientNetB0, ResNet50, and MobileNetV2 on waste item classification and proved that an ensemble methodology greatly enhanced the accuracy and strength of classification. Among standalone models, MobileNetV2 was the best compared to the rest, but the ensemble model resulted in the optimal overall performance, emphasizing the merit of using diverse architectures. Future development can target increasing the dataset size for improved generalization, investigating more sophisticated ensembling methods such as stacking and boosting, and model optimization for real-time deployment in intelligent waste management systems. Further development in data augmentation, class imbalance handling, and integrating explainable AI methods can also improve model strength. The combination of sensor-based multi-modal learning and model adaptation for IoT-based applications can make automated waste classification more feasible and effective for actual application.

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